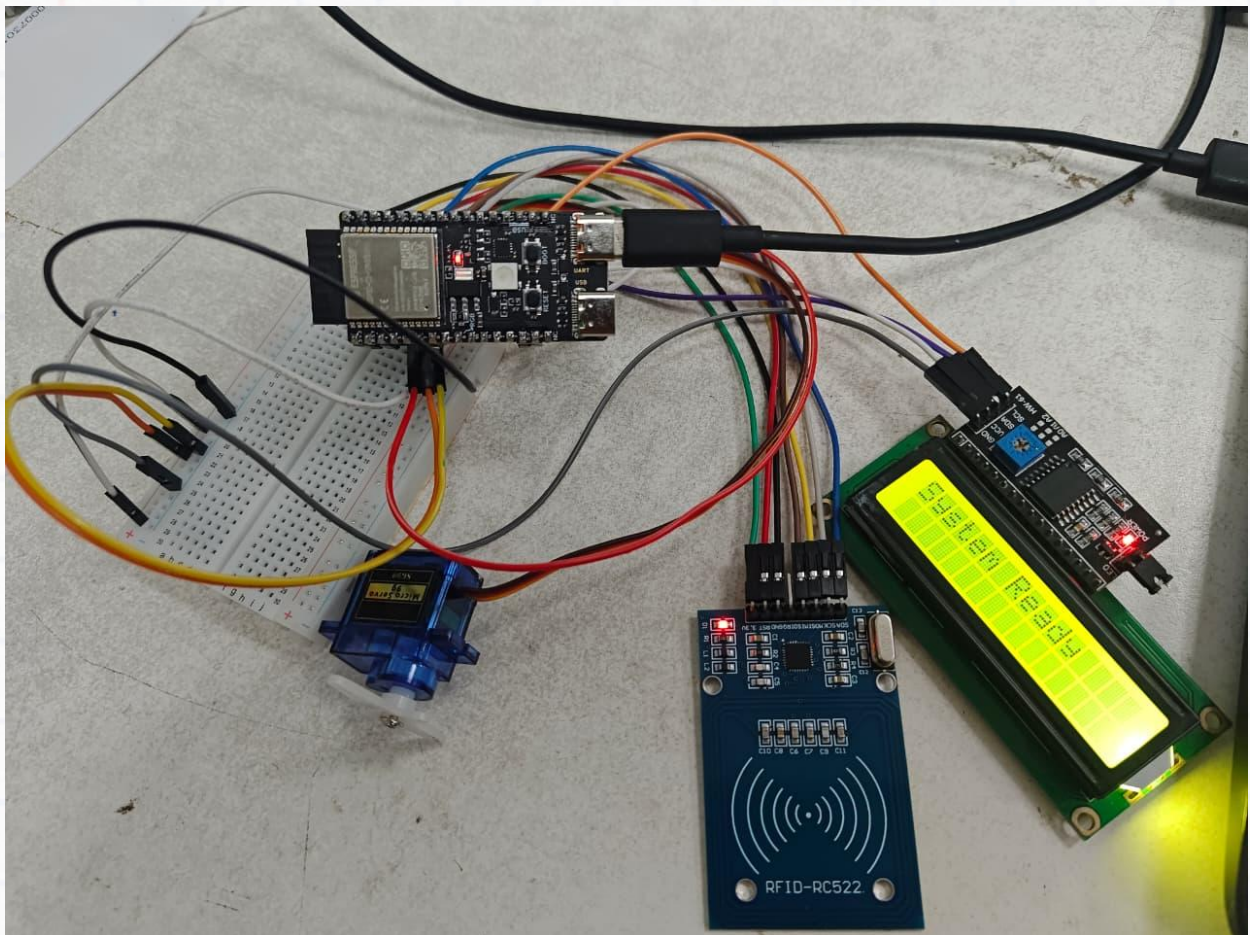


Build a RFID Fast Tag Reader



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Outcome

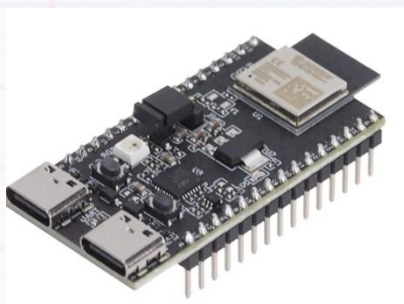
Build an RFID Fast Tag Reader that registers RFID card data, processes it, creates actions, and publishes the data to an online dashboard.

Concept

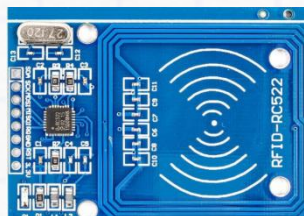
In this project, we mimic the working of a fast tag reader that reads RFID card data and operates the boom barrier in toll gates. Here we read RFID cards, verify them with the registry in the memory, open/close the gate, and finally send the data to the online dashboard.

Components

Sno	Components	Quantity
1.	ESP32 C6 Dev Module	1
2.	RC522 RFID Reader	1
3.	Breadboard	1
4.	Servo Motor	1
5.	16x2 LCD Display	1
6.	Jumper Wires	As needed
7.	USB to Type – C Cable	1



Esp 32 Module



RFID Reader



Breadboard



Servo Motor



16*2 LCD Display

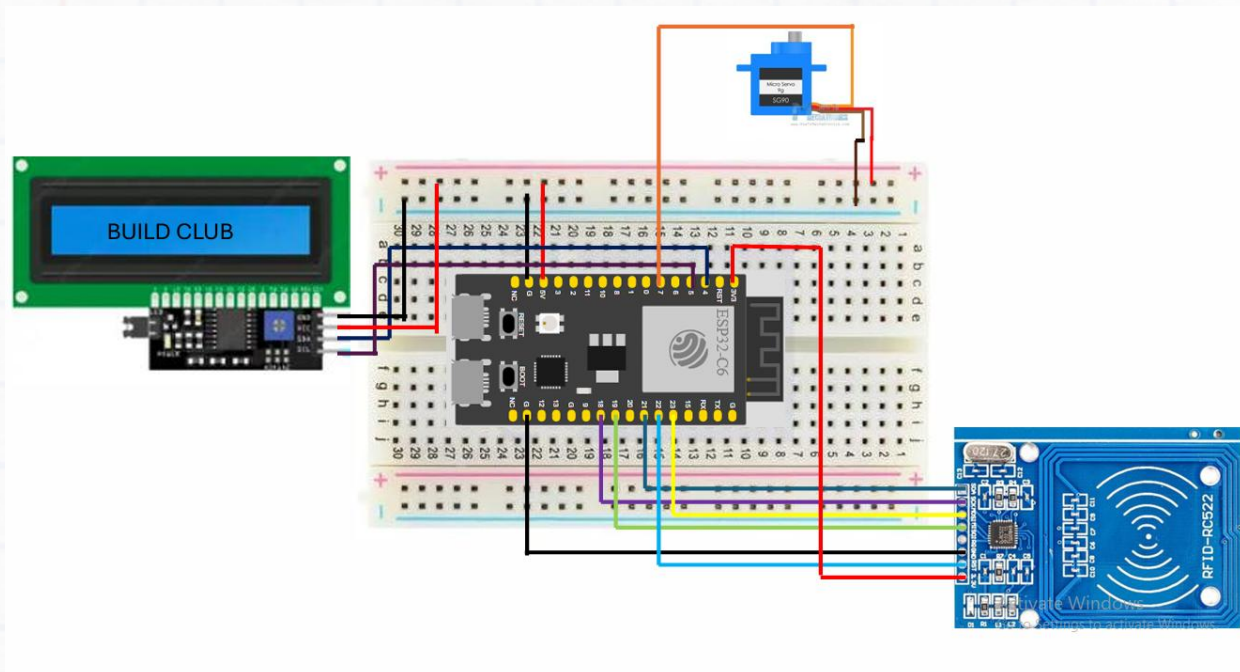


Jumper wires

Connections

Circuit Diagram

NOTE: Before starting the connections, verify that all the jumper wires are working using a multimeter. Also, ensure that the connections are strong, else the setup may not work.



Detailed Connection Steps

LCD with I2c Module:

- I2c GND to ESP32 GND
- I2c VCC to ESP32 V5
- I2c SDA to ESP32 G4
- I2c SCL to ESP32 G5

Servo

- Servo Out () to ESP32 G7
- Servo 5V to ESP32 V5
- Servo GND to ESP32 GND

RC522 Reader Module

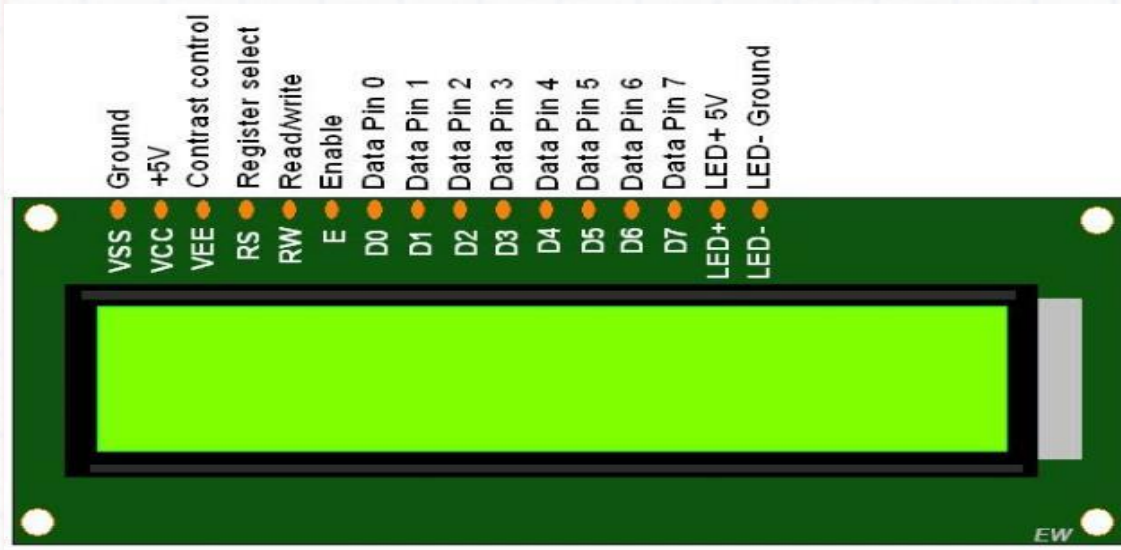
- SDA to ESP32 GPIO 21
- SCL to ESP32 GPIO 18
- MOSI to ESP32 GPIO 23
- MISO to ESP32 GPIO 19
- RST to ESP32 GPIO 22
- 3.3V to ESP32 3.3V
- GND to ESP32 GND

Install Required Libraries

- Go to Sketch > Include Library > Manage Libraries.
- Connect Your ESP32 C-6 Module
- Connect your ESP32 Module to your computer using a USB cable.
- Go to Tools > Board > ESP32C6 Dev Module.
- Go to Tools > Port and select the COM port to which your ESP32C-6 is connected.
- Go to sketch > Include Library > Manage Libraries.
- Search ESP32servo and install ESP32servo by kelvinHarrington.
- Search LiquidCrystal I2C and install LiquidCrystal I2C by Frank de Brabander.
- Search BlynkSimpleEsp32 and install by Khoi Hoang
- Search MFRC522 and install by GitHub community.

Part 1: Experimenting with the components

1. 16*2 LCD Display



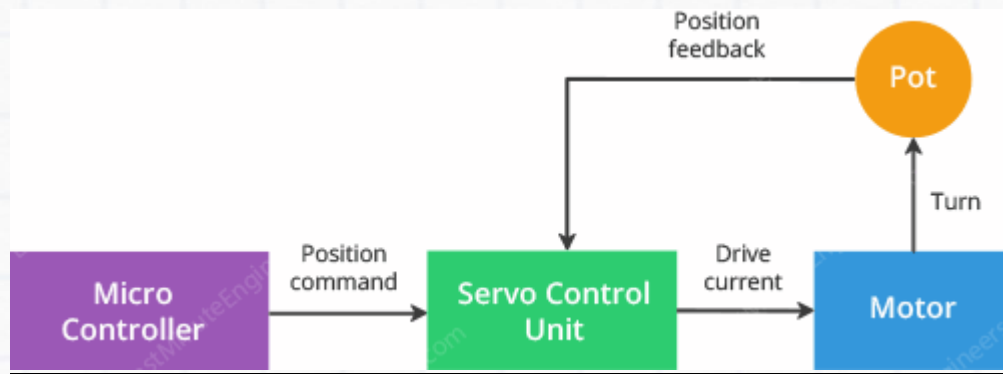
LCD (Liquid Crystal Display) is a flat panel display that uses liquid crystals to display information. It consists of an array of small segments called pixels, which can be manipulated to display status or parameters.

A 16*2 LCD can display 16 characters or numbers column-wise and 2 in row-wise.

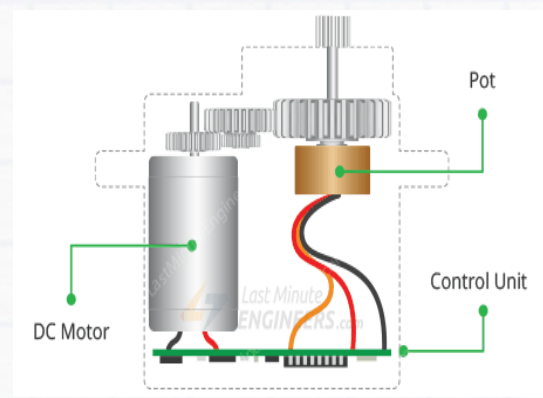
2. DC Servo motor

A servo system primarily consists of three basic components: a controlled device, an output sensor, and a feedback system. This is an automatic closed loop control system. Here instead of controlling a device by applying the variable input signal, the device is controlled by a feedback signal generated by comparing output signal and reference input signal. Any electrical motor can be utilised as a servo motor if it is controlled by servomechanism.

In the DC servo motor, a small DC motor connected to the output shaft through the gears. The output shaft drives a servo arm and is also connected to a potentiometer (pot). The potentiometer provides position feedback to the servo control unit where the current position of the motor is compared to the target position. According to the error, the control unit corrects the actual position of the motor so that it matches the target position

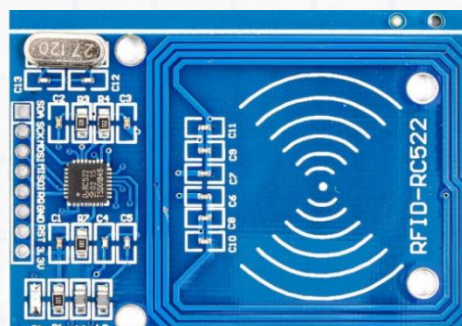


DC Servo motor wiring connection



DC Servo motor Internal Illustration

3) RC522 RFID Reader Module



The RC522 is a widely used RFID reader module that operates at 13.56 MHz. It communicates with microcontrollers using the SPI interface (Serial Peripheral Interface). It can read information from compatible RFID tags/cards and send the data to the microcontroller for processing. Each RFID card/tag has a unique ID.

Code:

“Open the .ino file which was attached in the folder and write a code by the below instruction.”

Main Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <ESP32Servo.h>
#include <EEPROM.h>
#include <WiFi.h>
#include <SPI.h>
#include <MFRC522.h>
// ----- BLYNK SETTINGS -----
#define BLYNK_TEMPLATE_ID      "TEMPLATE_ID"
#define BLYNK_TEMPLATE_NAME    "TEMPLATE_NAME"
#define BLYNK_AUTH_TOKEN       "BLYNK_AUTH_TOKEN"
#include <BlynkSimpleEsp32.h>
// ----- PIN CONFIG -----
#define SERVO_PIN 7
// I2C pins for ESP32-C6 (LCD)
#define I2C_SDA 4
#define I2C_SCL 5
// RC522 RFID (SPI)
#define SS_PIN 21
#define RST_PIN 22
#define MOSI_PIN 23
#define MISO_PIN 19
#define SCK_PIN 18
// ----- OBJECTS -----
LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo doorServo;
```

```

MFRC522 mfrc522(SS_PIN, RST_PIN); // Create MFRC522 instance
BlynkTimer timer;
// ----- ACCESS CONTROL -----
String allowedCards[] = { "53FCC129", "D1472A02" }; // Add allowed UIDs here
int totalEntries = 0;
int accessGranted = 0;
int accessDenied = 0;
// Wi-Fi credentials
const char* ssid = "ENTER_YOUR_SSID";
const char* password = "ENTER_YOUR_PASSWORD";
// ----- SETUP -----
void setup() {
  Serial.begin(115200);
  Wire.begin(I2C_SDA, I2C_SCL);
  lcd.begin();
  lcd.backlight();
  lcd.print("Connecting WiFi...");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    lcd.print(".");
  }
  lcd.clear();
  lcd.print("WiFi Connected");
  // Initialize Blynk (non-blocking)
  Blynk.config(BLYNK_AUTH_TOKEN);
  Blynk.connect(); // Connect to Blynk server
  // Initialize servo
  doorServo.attach(SERVO_PIN);
  doorServo.write(0);
  // Initialize RFID AFTER WiFi + Blynk
  SPI.begin(SCK_PIN, MISO_PIN, MOSI_PIN, SS_PIN);
  mfrc522.PCD_Init();
  // Timer to send data to Blynk every 5 seconds
  timer.setInterval(5000L, sendDataToBlynk);
  lcd.clear();
  lcd.print("System Ready");
}
// ----- LOOP -----
void loop() {
  Blynk.run();
  timer.run();
  if (!mfrc522.PICC_IsNewCardPresent()) return;

```

```

if (!mfrc522.PICC_ReadCardSerial()) return;
// Read UID
String cardID = "";
for (byte i = 0; i < mfrc522.uid.size; i++) {
  if (mfrc522.uid.uidByte[i] < 0x10) cardID += "0";
  cardID += String(mfrc522.uid.uidByte[i], HEX);
}
cardID.toUpperCase();
Serial.println("Scanned UID: " + cardID);
totalEntries++;
if (isCardAllowed(cardID)) {
  grantAccess();
  accessGranted++;
} else {
  denyAccess();
  accessDenied++;
}
}
// ----- FUNCTIONS -----
bool isCardAllowed(String cardID) {
  for (String uid : allowedCards) {
    if (uid == cardID) return true;
  }
  return false;
}
void grantAccess() {
  lcd.clear();
  lcd.print("Access Granted");
  doorServo.write(90);
  delay(1000);    // small delay to allow servo to move
  doorServo.write(0);
  lcd.clear();
  lcd.print("System Ready");
}
void denyAccess() {
  lcd.clear();
  lcd.print("Access Denied");
  delay(1000);    // small delay for user to see message
  lcd.clear();
  lcd.print("System Ready");
}
// ----- BLYNK DATA -----
void sendDataToBlynk() {

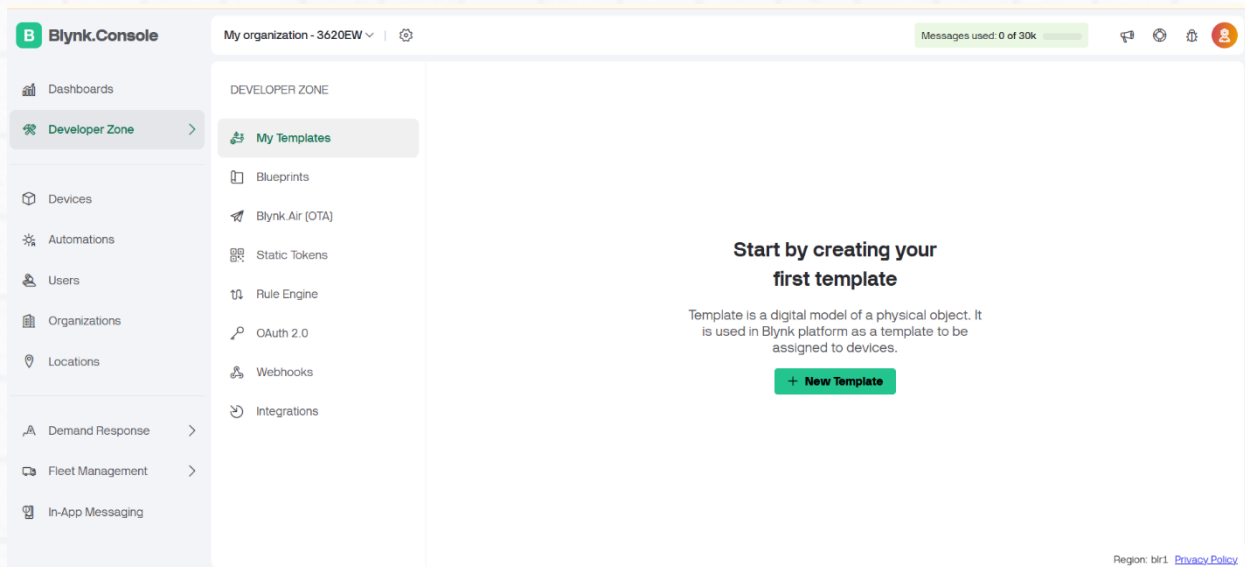
```

```
Blynk.virtualWrite(V0, totalEntries);
Blynk.virtualWrite(V1, accessGranted);
Blynk.virtualWrite(V2, accessDenied);
Serial.printf("Total: %d | Granted: %d | Denied: %d\n", totalEntries, accessGranted,
accessDenied);
}
```

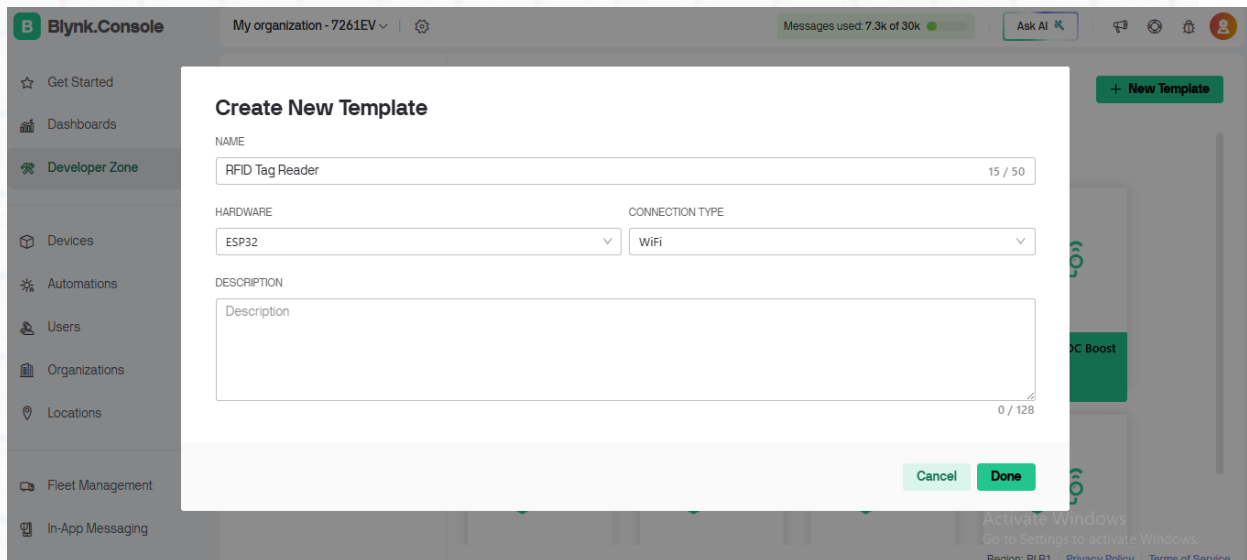
After completing these all the steps verify and upload a code to the esp32.

Setting up web dashboard – Blynk

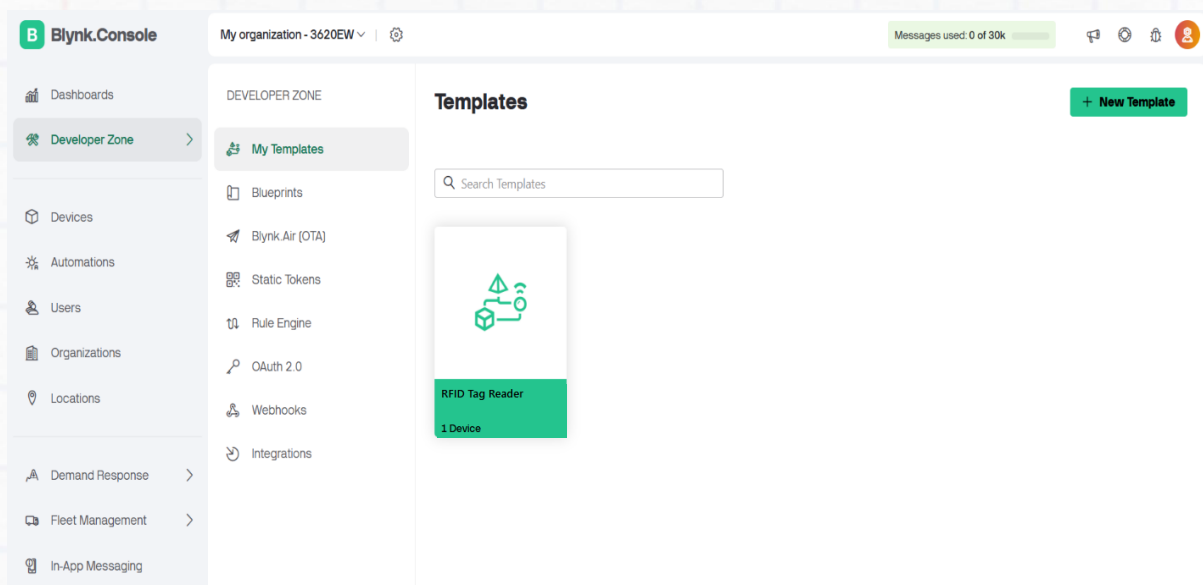
1. For monitoring, we will be using the Blynk IoT Platform. If you haven't, go to <https://blynk.io/> and sign up for free.
2. On the blynk website, click on Developer Zone >> My templates as shown in the below.



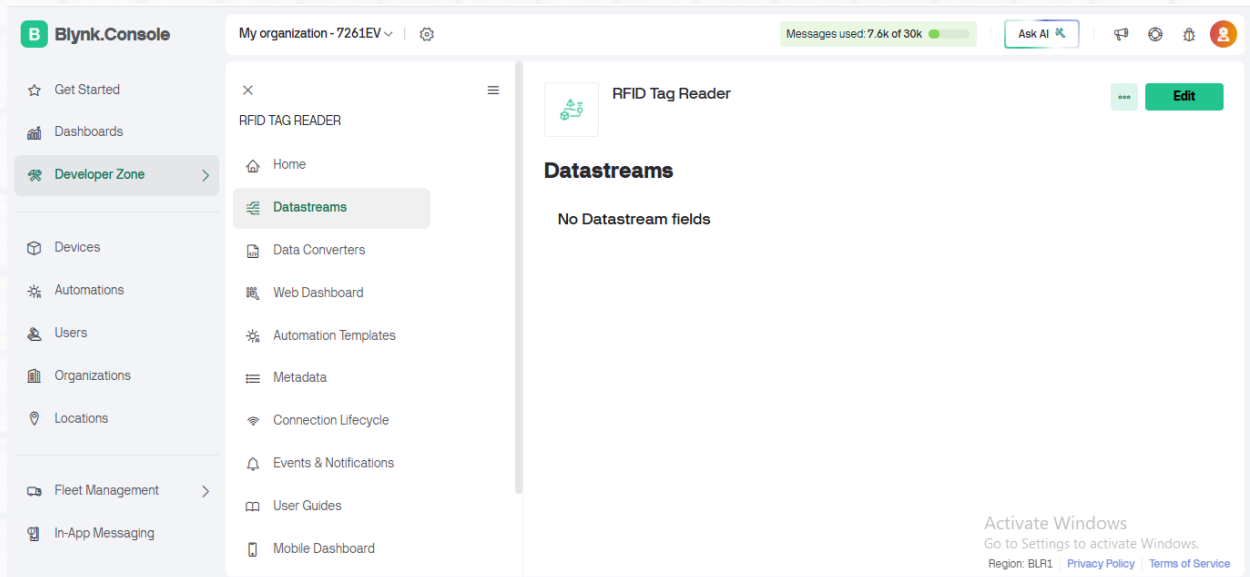
1. Select **New Template** and give the Name of the template as **RFID Tag Reader** and check once the hardware is **ESP32** and the connection type is **WIFI** and click done.



2. Now click on Developer Zone >> My templates >> RFID Tag Reader

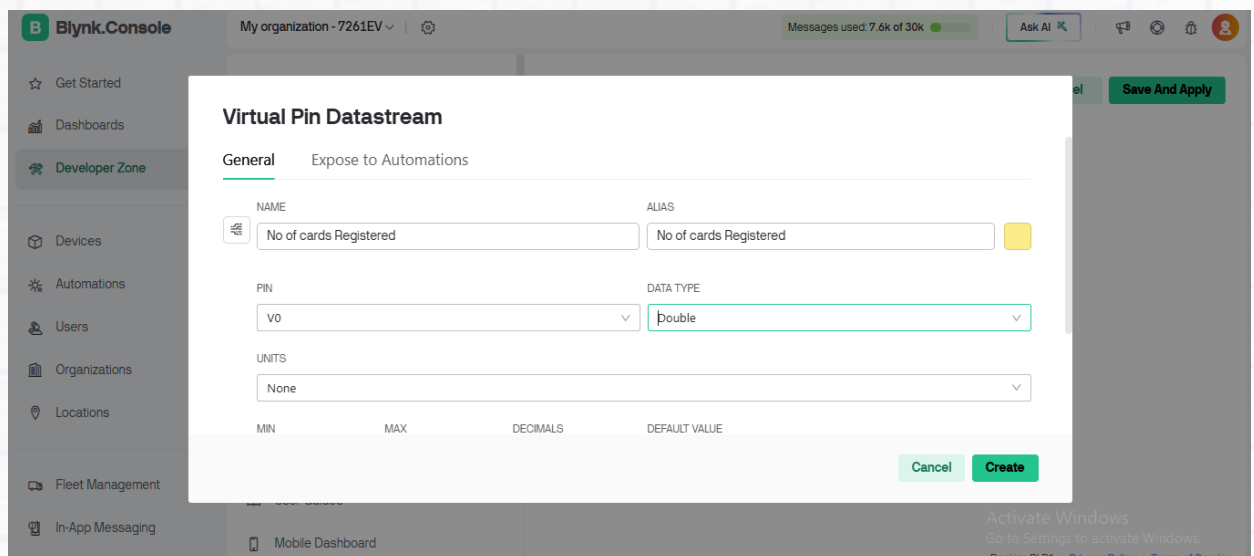


3. Now click on RFID Tag Reader >> Data streams. After this click on Edit option on the top of the right corner.

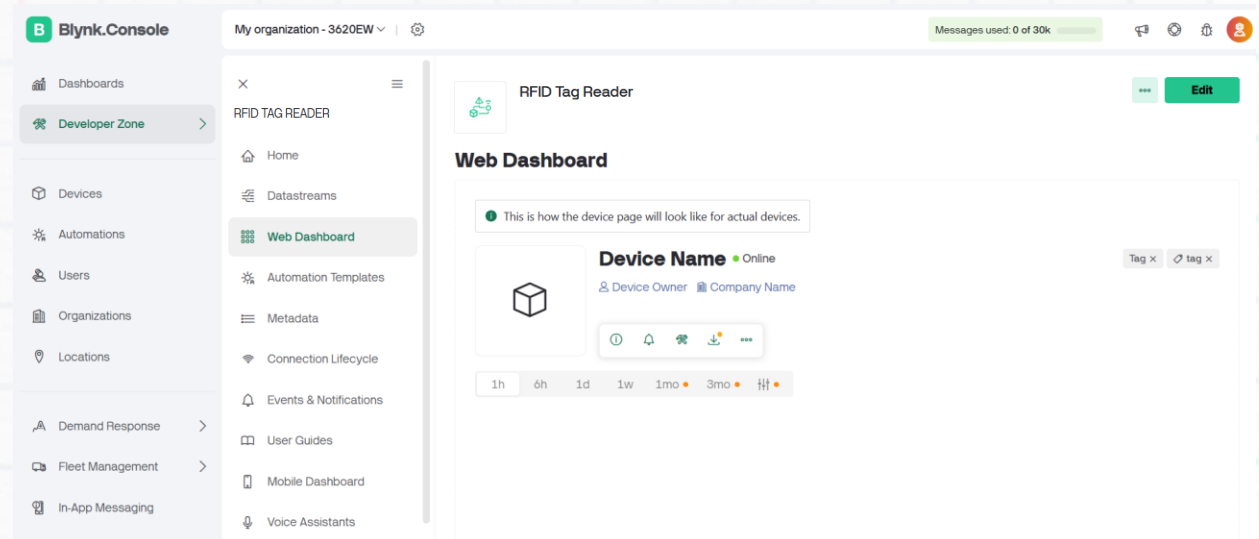


4. Now click New DataStream >> virtual pin. Give parameters Name as “No of cards Registered”, Pin as v0, Data type as Double and give then click create

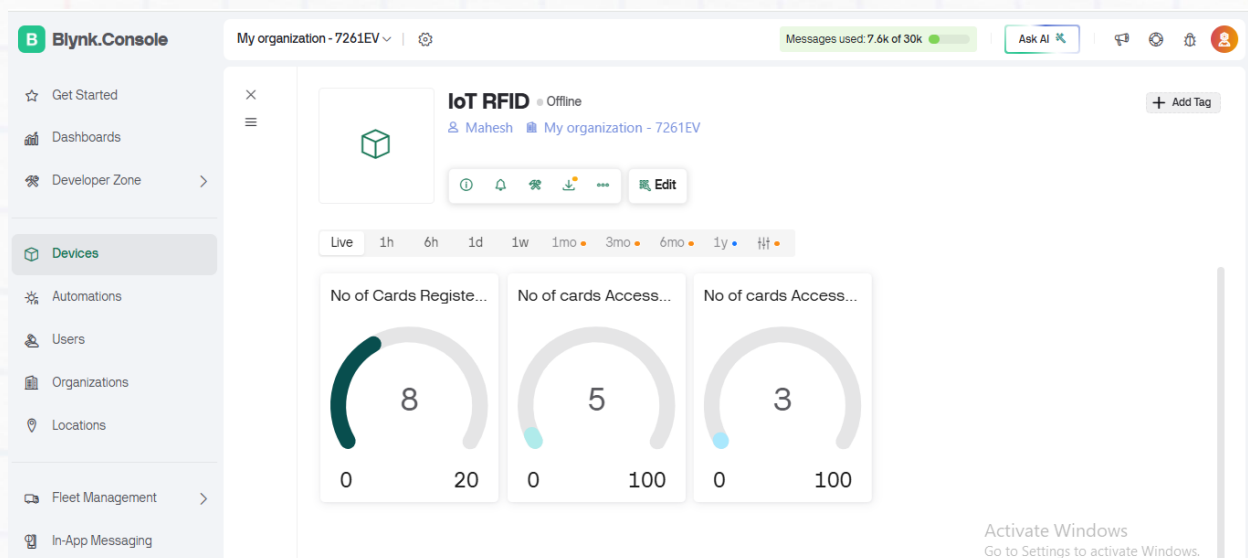
Note: Please Do the same thing for remaining parameters like No of cards Accessed and No of Cards Declined.



- After that click Developer zone >> My templates >> RFID Tag reader >> Web Dashboard on the left side.

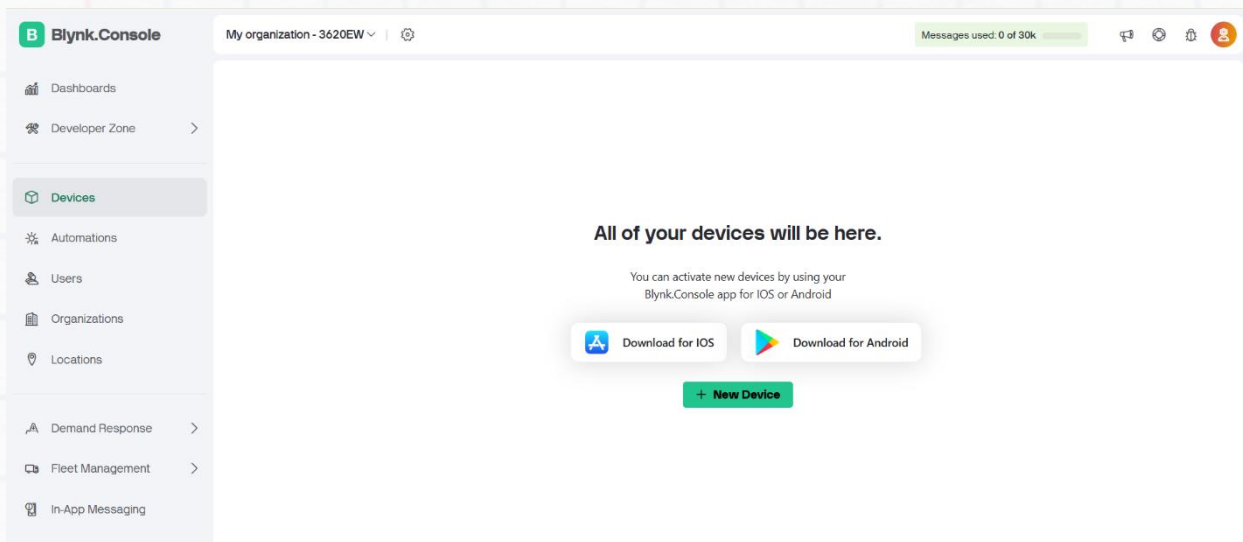


- Click Edit option in the top of the right side and scroll down the widget box double click Gauge.
- Click the gear icon (⚙️) in the gauge and give the title as No of cards Registered and data stream as No of cards Registered and save it.

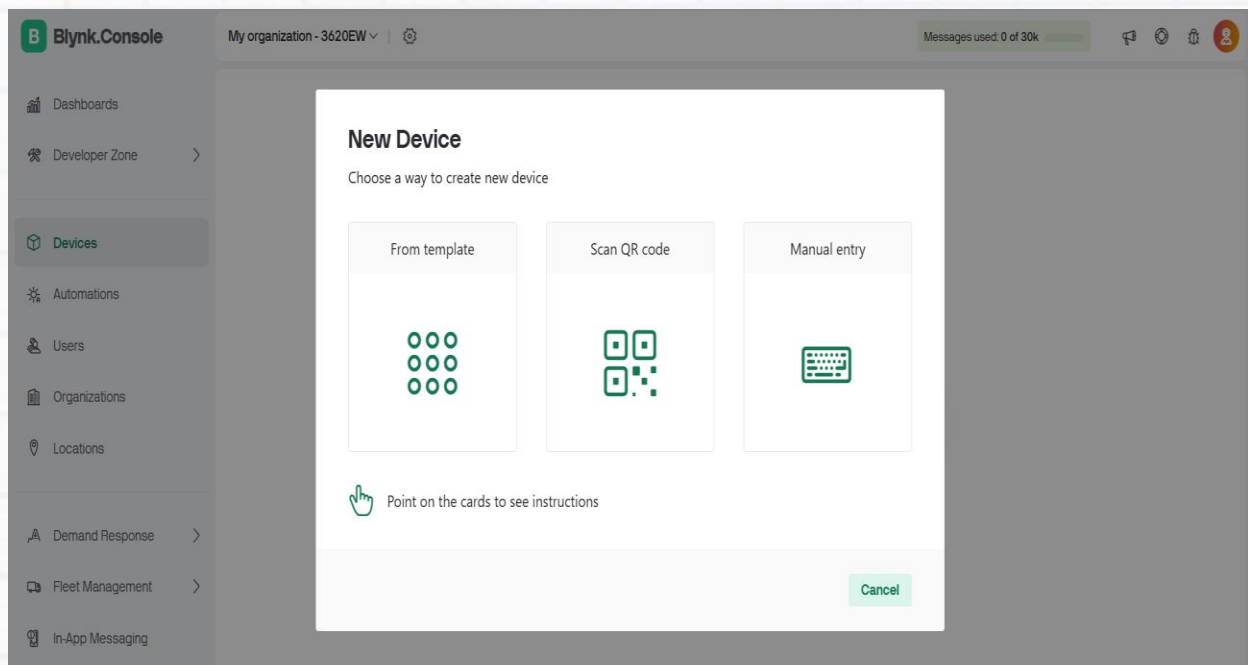


- Edit the gauges for remaining parameters and Click save and apply.

9. Now Click Devices and Click New Device.



10. Click New device and select From Template



11. Now Click Choose device >> RFID Tag Reader and click Create

12. Now everything is done, and you got BLYNK_AUTH_TOKEN Copy that Auth token on the top right part of the page and paste it into the BLYNK_AUTH_TOKEN variable, which is immediately below the Wi-Fi variables in the Arduino code.

Task

1. After erasing the current memory. Take 3 new RFID cards, add 2 cards. Keep one card unregistered. Check how the device is behaving with the 3 cards. Check whether the data is received on the Web dashboard.
2. Make a video presenting the build of RFID reader, how you completed the activities and challenges. Share it with the Build Club Community on the Discord Server!